

Databases, Dashboards & Death to Paper

School of Data LA — 2019/02/01

Data & Technology Services  Austin Transportation

Howdy!

Install Stuff?

github.com/cityofaustin/school-of-data

What we'll cover

- Our Team
- Our Smart City Vision
- Software/Platform Choices
- Systems Architecture & Project Examples
- Hands-on workshop
 - Python ETL
 - JavaScript DataViz

Our Team

What does Austin Transportation do?

Operate traffic and pedestrian signals

Issue parking, right-of-way, and dockless mobility permits

Design and engineer the transportation system

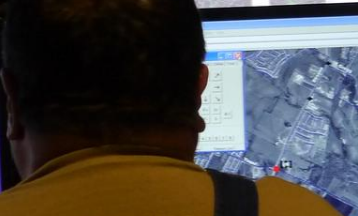
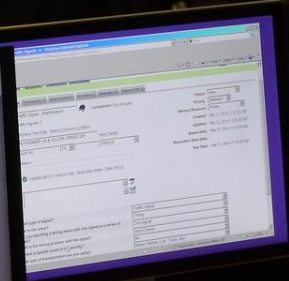
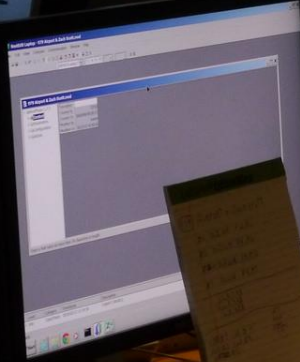
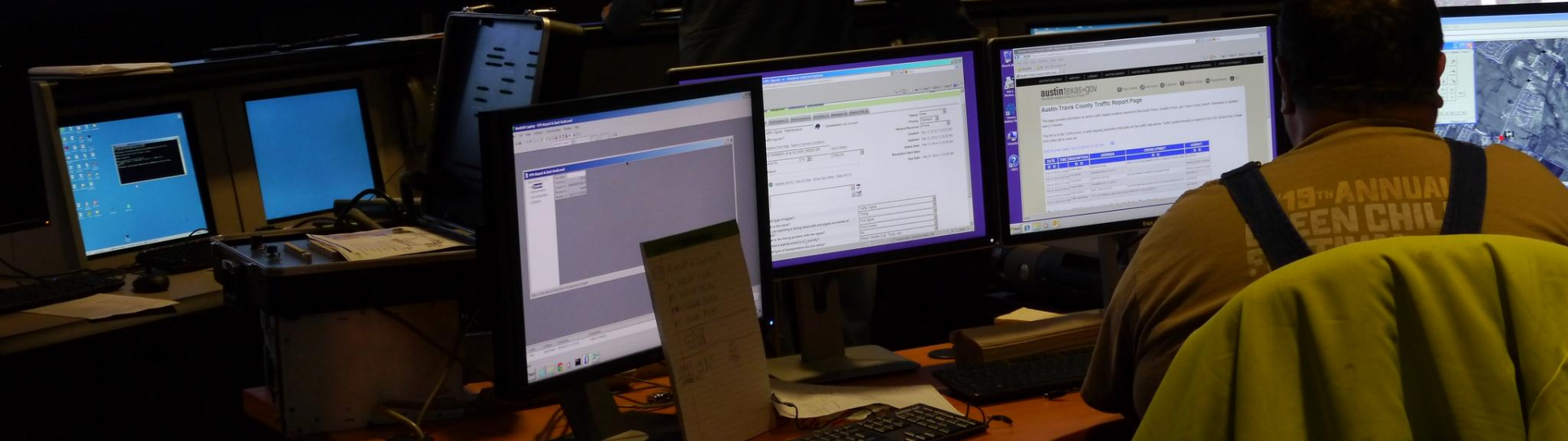
Install signs and pavement markings

...and more



VOLUMES

SIGNAL
TIMING



Data & Technology Services

Mission:

We use technology to solve real problems for real people related to mobility in Austin.

Objectives:

- We publish reliable and authoritative data and information.
- We support real-time performance measurement.
- We empower staff and residents to find information easily.
- We position our team to adopt new technologies quickly.

Our Team



John Clary
IT Supervisor Sr



Scott Reedy
IT Business System Analyst Sr



Diana Martin
IT Business System Analyst Sr



Garry Olson Jr
IT Business System Analyst



Tracy Linder
Business System Analyst Sr



Mateo Clarke
Software Developer Sr



Alan De Anda
IT Geospatial Technician



Matt Stevens
IT Geospatial Technician



Zhaowei "Joey" Liang
Software Developer Intern

Our Services

Geospatial

Print Maps

Web Maps

Spatial Analysis

**Data Creation +
Design**

App Support

App Support

App Configuration

**Process
Refinement**

**Data Creation +
Design**

Infrastructure + Engineering

**Desktop Hardware
+ Software**

**Systems
Integration**

Data Engineering

Procurement

Professional

**Project
Management**

**Solution
Consulting**

**Acceptance
Testing**

Reporting

OPEN BY DEFAULT

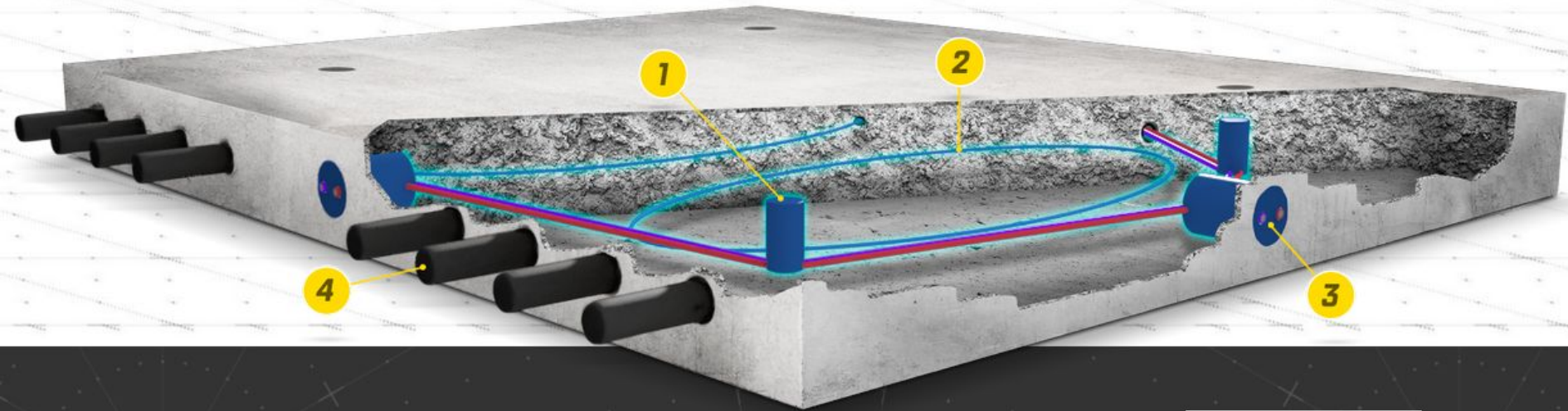


"It'd be a lot cooler if you did."

Our Smart City Vision

So do you big data?

Smart Pavement



<http://integratedroadways.com>

**City of Austin Traffic Signal Division
Signal Trouble Ticket**

Location <u>71 @ CENTRAL OF THE HILL</u>	
Intersect # <u>430</u>	APD Case #
Date <u>2-1-17</u>	Weather <u>SUNNY</u>
Source <u>213</u>	
Work Location: <u>SG</u>	Work Type: <u>PM</u>

Time Received
Time Arrived @ Loc. <u>9:00</u>
Time Departed Loc. <u>12:00</u>
Time Back @ Shop
Time Back @ Home

Task Order # 2423257000

CSR # no.

Trouble Reported / Work Assigned: <u>PM TXDOT</u>	Storm:	Y	N
	Replace Controller:	Y	N
	Controller Ver.:	1.4X	1.7X
	Controller Brand:		
	Loops:	Y	N
Problem Found: <u>EB AMBER OUT, WB 5 SEC MESALEGA</u>	Video:	Y	N
<u>Q3 & Q4 BAD LEAD-IN CABLE</u>	CMU Fault:		
<u>EB Q2 SET BACK LOOP PULL BOX -</u>	UPS Status:	N/A	Y
<u>PB COVER DAMAGED</u>	Pre-empt Working:	N/A	Y
<u>WB SEGMENT CABLE HANGING DOWN</u>	QTY	Materials Used	

STATUS OF DETECTION REPAIRS
CITY OF AUSTIN, TRANSPORTATION
TRAFFIC SIGNAL DIVISION

Legend:

	No Issues
	Repairs underway
	Under Repair ATS
	Scheduled for Final Tie In
	ATD Crew to Do w/o Others
	In CBD
	In CBD or no Detection (pretimed)
	Locate Pullbox
	Identified before 2014
	Unchecked
	Complete

DATE 12/30/16

Location	Current Needed Repairs	Most Recent Research By	Date Identified	Status	Comments	#
Guadalupe	NB setback Loops bad - 3 sets of 2	Willard	06/30/15			
amar @	EB Video Detection Bad	Nazlie	11/02/16			
amar @					Need EB Loops Cut. SB Cut	
NorthLoop	SB LT Loop BAD	Willard	06/30/15		7/30/15	

 Detection_Master - Copy.xlsm	129,376 KB
 Detection_Master_SHIFT.xlsm	129,376 KB
 Detection_Master_REAL.xlsm	118,977 KB

Smart city vision

Let's use technology to solve real problems for real people



Maturity Map

Use this map as a jumping off point for talking about your team or department's work styles, technology in use, and social engagement.

ENGAGE PEOPLE

How do you reach out to people most often?



1-Way/Broadcast
Engagement



Receive Input/
2-Way Engagement



Close the
Feedback Loop



Inclusive Partnerships/
Relationships

LEAD COLLABORATIVELY

What does leadership look like on your teams?



Hierarchical



Encourages Team
Problem Solving



Shares Accountability
of Outcomes



Transparent Leadership/
Citizen-Driven Transformation

WORK ACROSS DISCIPLINES, SYSTEMS

How do teams work together?



Silos



Across Disciplines



Across Teams
or Units



Across Systems

TECH & DATA

How is content managed and technology used?



Paper Trial/
No Data



Some Automation/
Some Data



Transactional
Services/Analytics



Integrated
Systems

INTERFACE

How evolved is your user experience?



In-Person
Content Exchange



Basic Digital
Interface



Responsive
Site/Content



Anticipate Needs/
Proactive Transformation

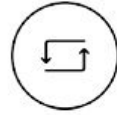
Tech & Data Maturity Map



Paper Trial/
No Data



Some Automation/
Some Data



Transactional
Services/Analytics



Integrated
Systems

Sprint Reviews



Accountability

- Deliver updates
- Demo work in progress

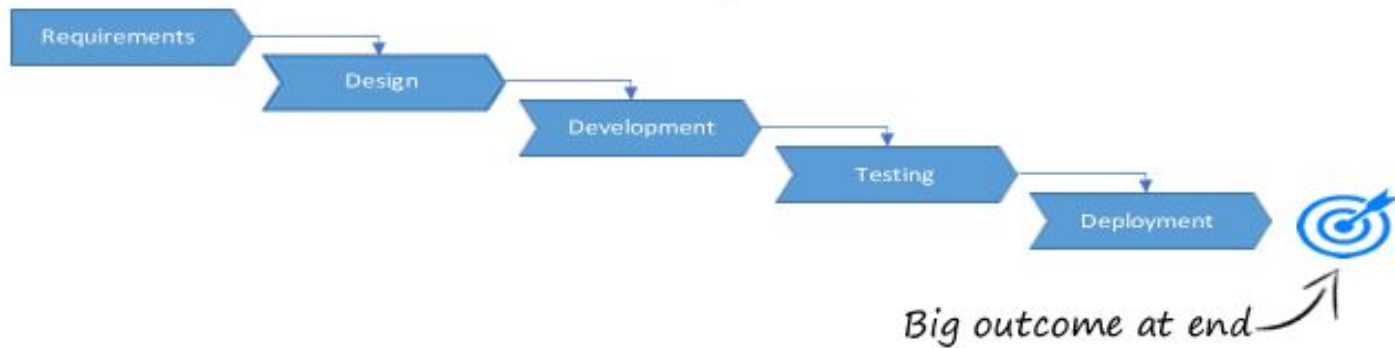
Better End Product

- Gather frequent feedback
- Adjust priorities and update backlog

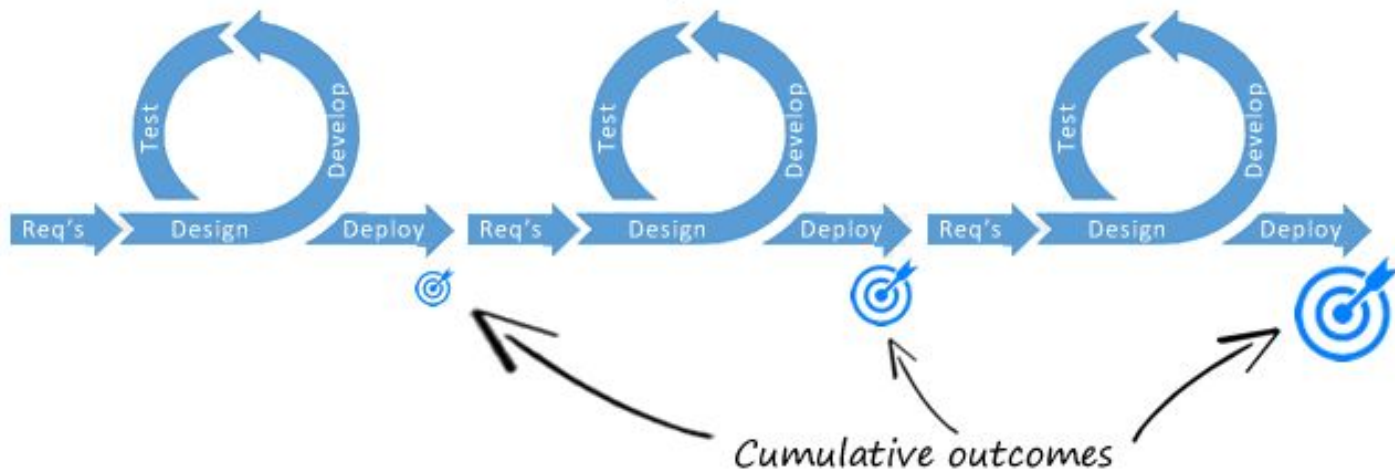
Better Team

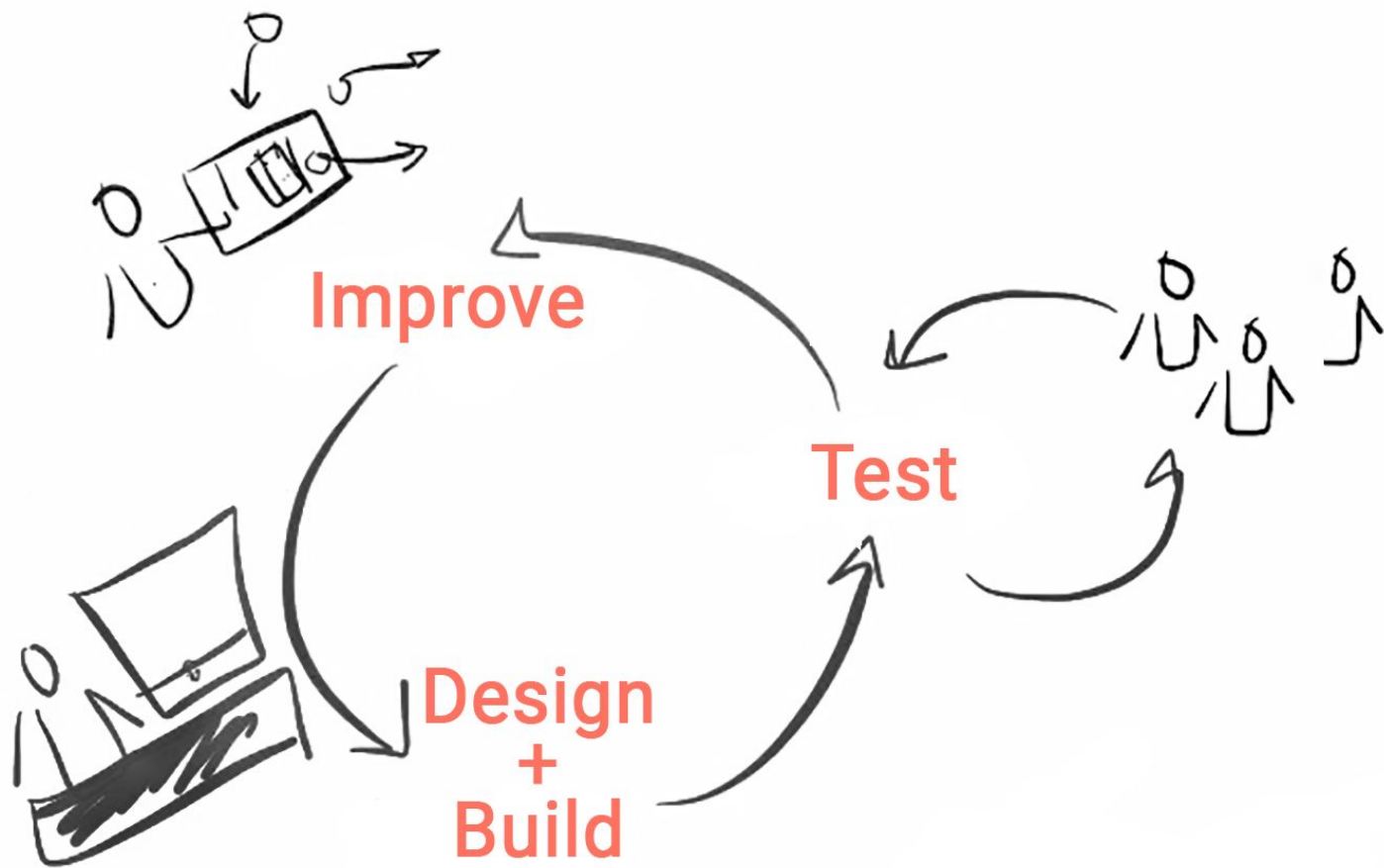
- Share insights and learnings
- Reflect on what went well and where we can improve

Waterfall



Agile





Software/Platform Choices

IT Bureaucratic Context

Centralized IT does some things reliably well:

- Phones & teleconferencing
- Email
- Networking
- CISO/Security
- Training

But... many of our dept IT needs are **domain-specific** and/or **time-sensitive**:

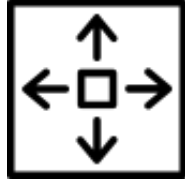
- Data Pipeline/ETL
- Custom reporting
- Custom software/web development
- Project management & user research

Tooling choices

- Tooling is emblematic of the space(s) we come from:
 - **Open-source civic tech** projects vs top down enterprise IT
- A little bit of coding goes a long way
- **Avoiding enterprise licenses**
 - Especially when costs are tied to the number of users



Criteria for platform/open source software choices



Customizable

Expandable

Portable

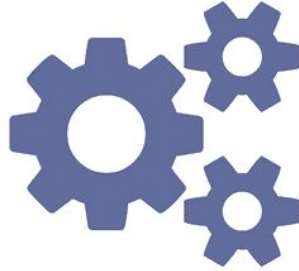
Supported

Is this too much to ask for?



Database

+



Business
Rules

+



Web Forms

No/Low Code App Development

Online platforms that let you build custom business applications (forms and pages) writing little to no code.



ArcGIS Online

Why We Chose Knack

Out-of-the-Box Spatial

Support for latitude / longitude, geocoding, and location services

Extensible

Custom JS, CSS

Plays Well With Others

Active Directory, E-Payments. Well-documented REST API.

Affordable

Tiered subscription levels. Unlimited Users is standard.



GIS



**Open Data
Portal**



**Service
Requests**



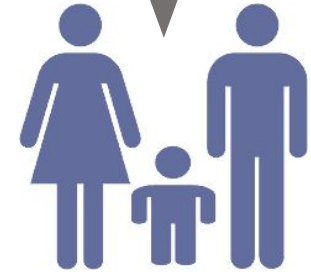
**Performance
Hub**



**Transportation
Staff**



**Business
Apps**



The Public



16 spreadsheets retired

100k+ digital records created

70 active users

Less than **\$10,000/yr**

Integrated with 311, GIS,
Open Data Portal*

Systems Architecture & Project Examples

Purchase Requests



Finance and Purchasing

Austin Transportation Department

[Home](#) [Purchase Requests](#) [Finance & Purchasing](#) [Help](#)

Logged in as **John Clary** - [Account Settings](#) - [Log Out](#)

All Purchase Requests

[All Items](#) [Not Submitted](#) [Waiting for Approval](#) [Finance Review](#) [Returned](#) [Rejected](#) [Closed](#) [Cancelled](#)

search by keyword

Showing 1-25 of 285

Page 1 of 12

Details	ID	Request Status	Subject	Total Cost	Requester	Reviewer/ Approver	Division	Submitted Date	Modified Date
	2019-0279	Processing Purchasing	35th & Pecos Fiber	\$4,618.64	Brian Craig	Ramona Aguilar	ARTERIAL MANAGEMENT	01/16/2019	01/29/2019 2:29pm
	2019-0280	Budget Review	pipe threader parts	\$81.45	Billy Bolander		ARTERIAL MANAGEMENT	01/16/2019	01/29/2019 2:23pm
	2019-0260	Pending Invoice	Span Wire Brackets for hanging overhead street name signs.	\$2,094.38	Kenny Scott		SIGNS AND MARKINGS	01/10/2019	01/29/2019 2:21pm
	2019-0328	Waiting for Approval	Video Imaging Vehicle Detection System	\$126,225.00	Ray Flores	Brian Craig	FINANCE AND ADMINISTRATION	01/29/2019	01/29/2019 2:20pm
	2019-0256	Pending Invoice	Park Bend & Mopac Fiber Repair	\$7,418.32	Brian Craig		ARTERIAL MANAGEMENT	01/09/2019	01/29/2019 2:13pm
	2019-0326	Purchase Review	Tablet and Accessories Purchase (Barton Oaks Reception)	\$184.89	Garry Olson Jr.		DATA AND TECHNOLOGY SERVICES	01/29/2019	01/29/2019 1:43pm

purchase_requests Fields Records Tasks Settings

[view live app](#) 

PAGES

ADD  Text ▾  123 Number ▾  Choice ▾  Date ▾  File ▾  Personal ▾  Other ▾

+ ADD

reviews

miscellaneous_payments

+ ADD

Admin

Assigned To



Purchase Review



Budget Review



Purchasing Comments



Process Purchase Request



Accounts Payable Process



MODIFIED DATE



Submitted Date



AMOUNT PAID



[+ Add Connection](#)

purchase_requests connects to ▼

 MODIFIED_BY

Commodity Description

 [Contract Agreement](#)

 Vendor Contact Information

 inventory_item

Connects to purchase_requests ▼

 purchase_request

Request ID



Common Requirements for Admin & Back-Office

- Email notifications
- Approval workflows
- Search, sort, export data
- Single sign-on authentication & role-based authorization



Work & Asset Management



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1029: 830 BLK W RUNDBERG LN

Little Walnut Creek Library, HEB



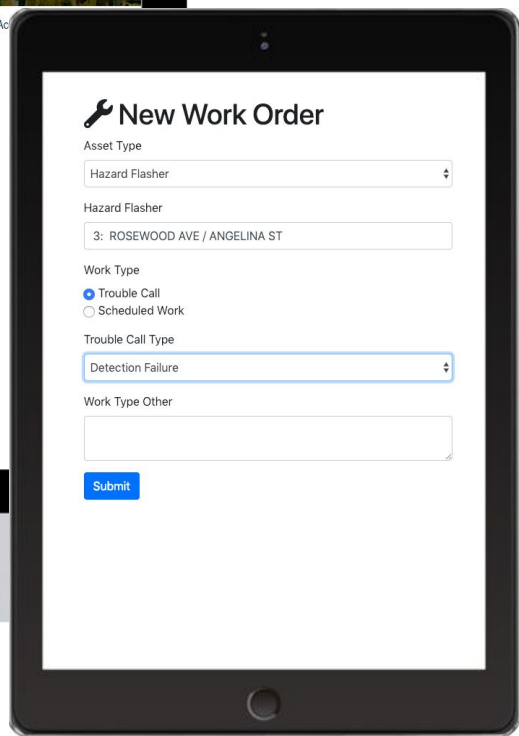
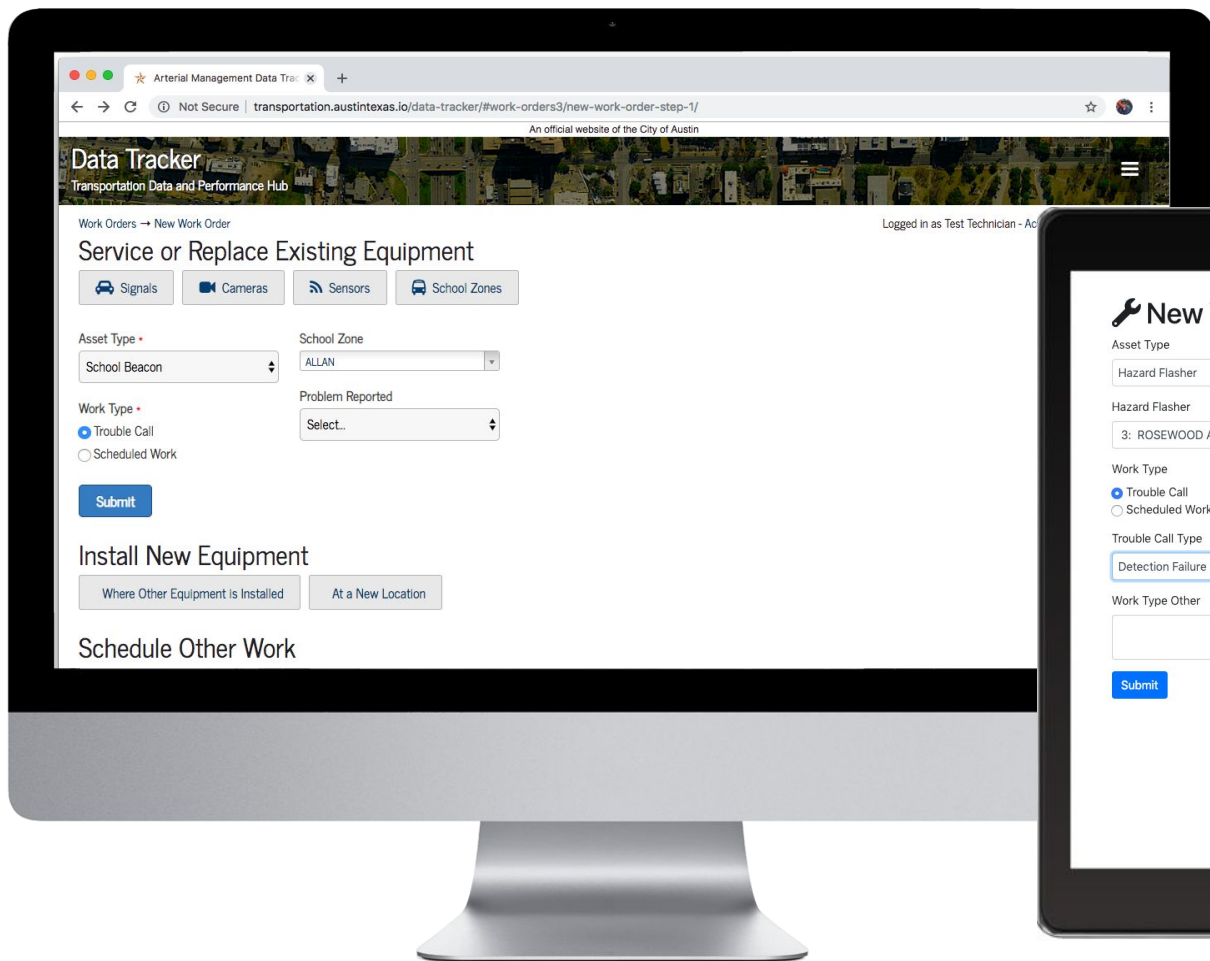
[View on Google Maps](#)

[Edit Signal](#)

[Edit Detection](#)

Signal ID	1029
Type	TRAFFIC
Status	TURNED_ON
Control	MASTER
Comment	Converted from PHB ID 4020
Modified	02/14/2017 10:49am
Modified By	John Clary

Turn On Date	11/16/2016
Owner	COA
Map	Asset Map

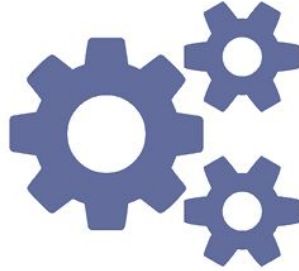


knack*



Database

+



Business
Rules

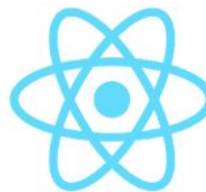
+



Web Forms

knack*

+

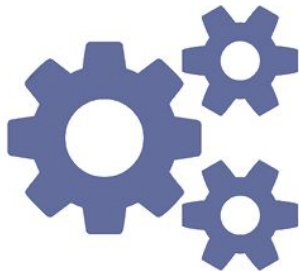


React



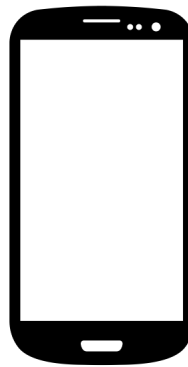
Database

+



Business
Rules

+



Single
Page App

Data & Performance Hub
transportation.austintexas.io

An official website of the City of Austin

Data and Performance Hub

City of Austin Transportation Department

Signal Operations

Traffic Signal Monitor



Signal Projects



Signal Timing



Signal Requests



Device Status



Operations Overview



Traffic Volumes (Beta)



Operations Map (Beta)



Maps and Resources

Signal Asset Map



Preventative Maintenance



Data Tracker



Open Data

Dockless Mobility Data



Open Data Inventory



City of Austin GitHub



Signal Requests

Transportation Data and Performance Hub

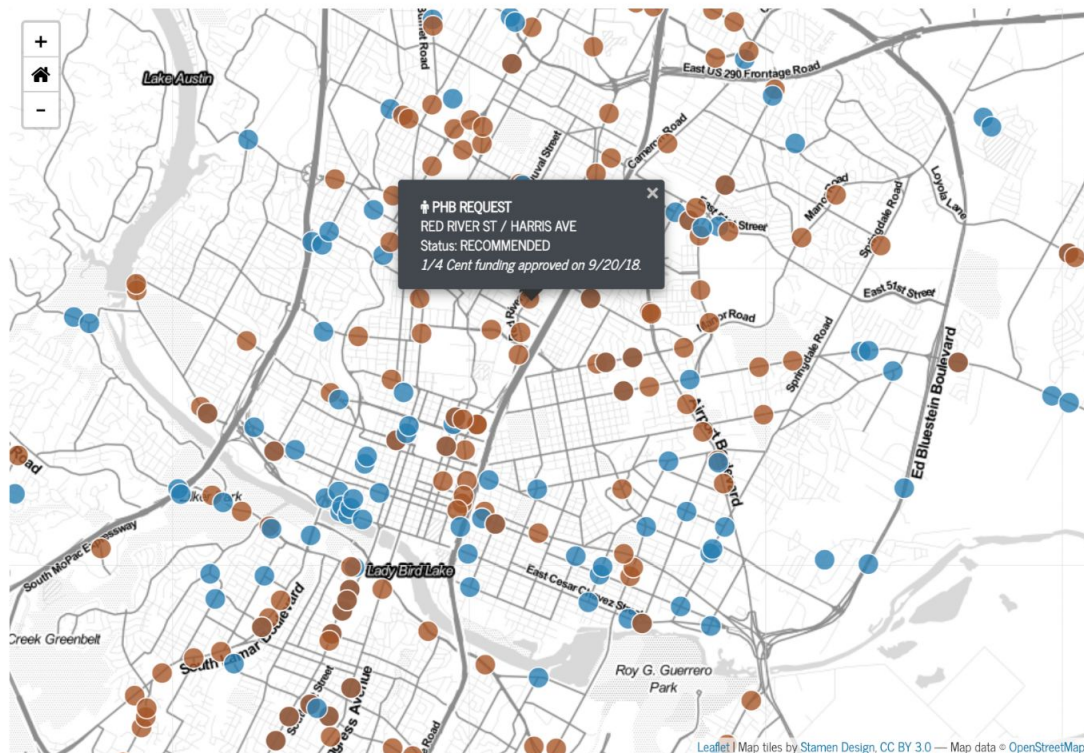
Find a Signal Request

Search by street name or status...



Location	Type	Status
11TH ST / BRANCH ST	TRAFFIC	UNDER EVALUATION
12TH ST / LUNA ST (Givens Park)	PHB	TURNED_ON
13TH ST / SAN JACINTO BLVD	TRAFFIC	UNDER EVALUATION
13TH ST / SAN JACINTO BLVD	PHB	UNDER EVALUATION
16TH ST / SAN JACINTO BLVD	TRAFFIC	UNDER EVALUATION
18TH ST / NUECES ST	TRAFFIC	RECENTLY RECEIVED
1ST ST / 1ST ST (Akins High School - Main Entrance)	TRAFFIC	UNDER EVALUATION
1ST ST / 1ST ST (Mary Moore Searight off leash trail)	PHB	UNDER EVALUATION
1ST ST / BRAMBLE DR	PHB	RECENTLY RECEIVED
1ST ST / CAMPERDOWN ELM DR	PHB	TURNED_ON
1ST ST / COPELAND ST (Texas School for the Deaf)	TRAFFIC	NOT RECOMMENDED
1ST ST / COPELAND ST (Texas School for the Deaf)	PHB	NOT RECOMMENDED
1ST ST / DESERT PRIMROSE DR	PHB	CONSTRUCTED

Showing 1 to 643 of 643 entries



Socrata SODA API + Jekyll + Github Pages

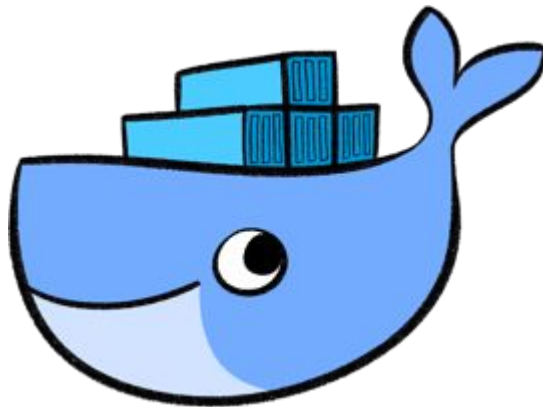


GitHub

jekyll 

Docker

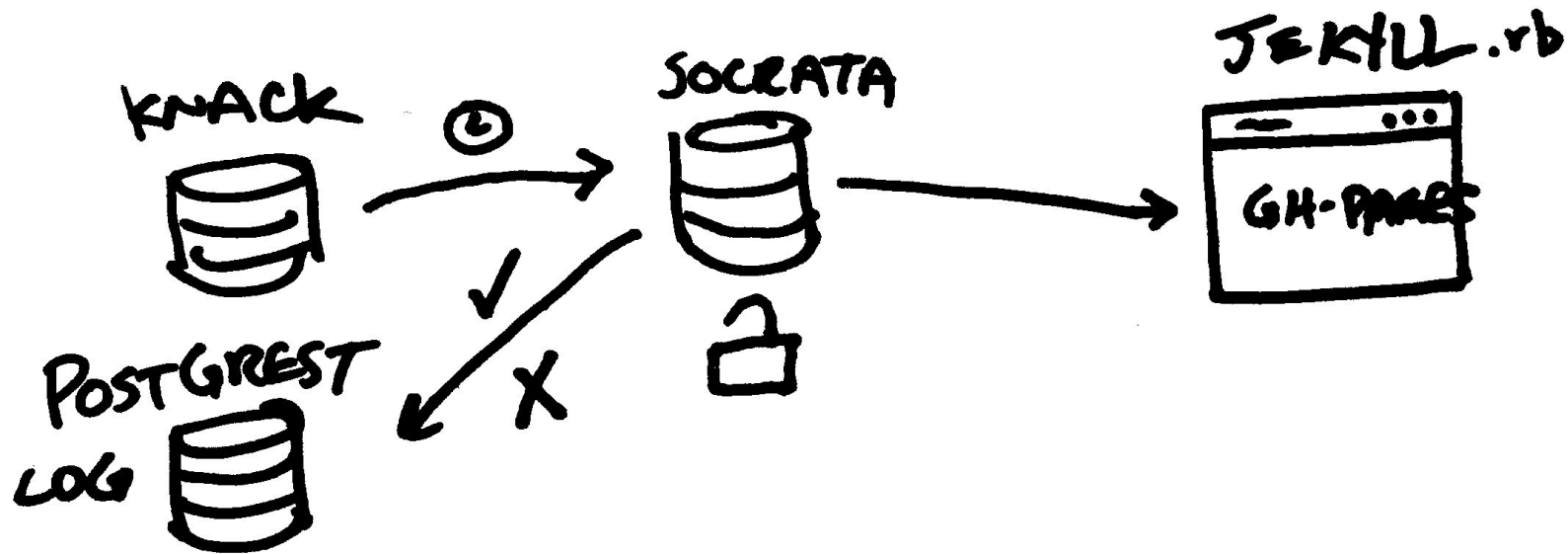
- Docker is a computer program that performs operating-system-level virtualization, also known as "containerization"
- transportation-data-publishing



PostgREST + PostgreSQL

- RESTful API from any PostgreSQL database
- Faster API than you are likely to write from scratch.





Dockless Mobility
dockless.austintexas.io

<> Code

! Issues 40

🔗 Pull requests 10

📁 Projects 0

📊 Insights

A data standard for Mobility as a Service Providers who work within in the City of Los Angeles

📦 225 commits

🌿 12 branches

📦 6 releases

👤 30 contributors

📄 CC0-1.0

Branch: dev ▾

New pull request

Create new file

Upload files

Find file

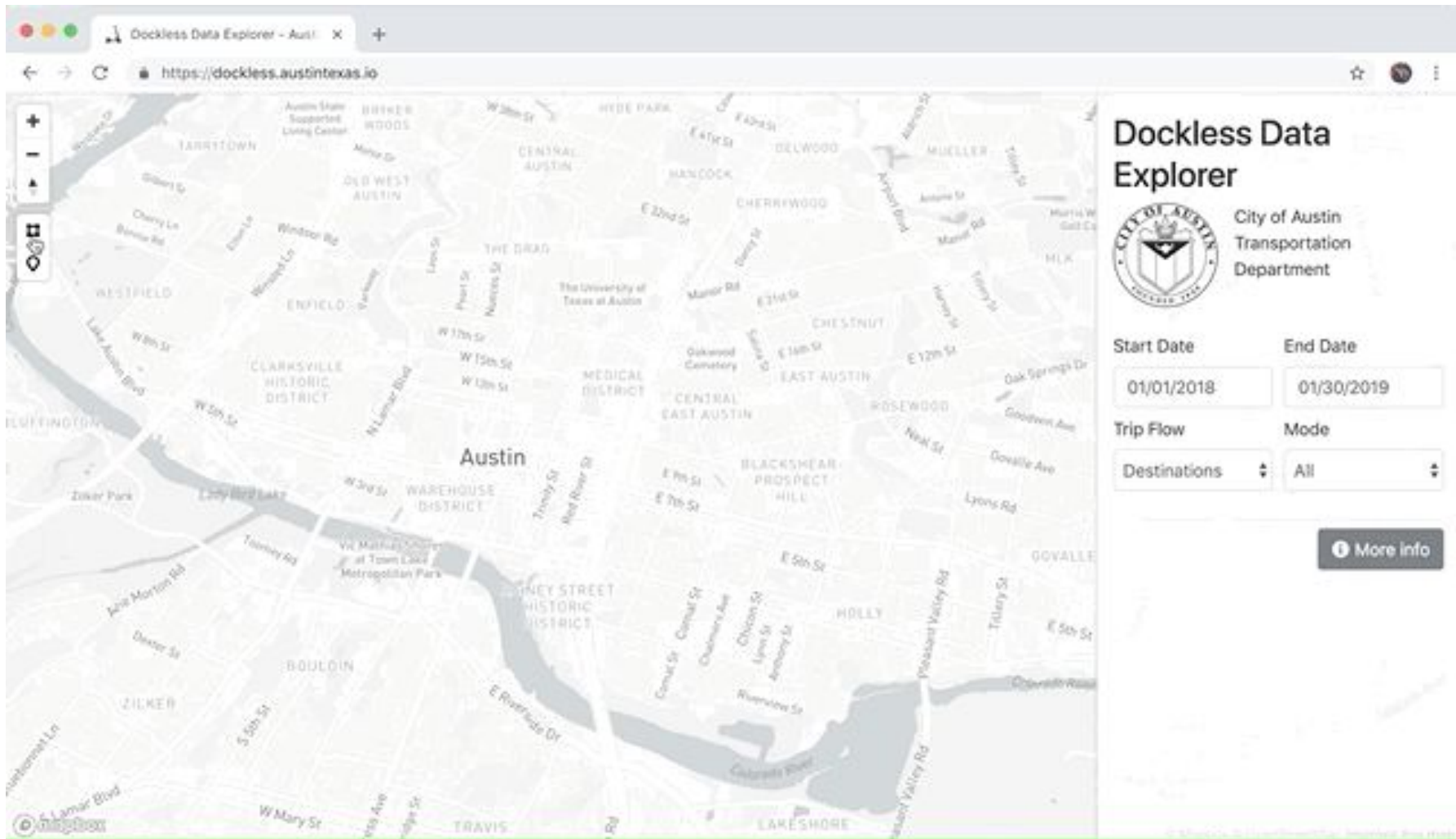
Clone or download ▾



asadowns and hunterowens Add GBFS Feed Url to providers.csv (#205) ...

Latest commit 1ced6b9 8 days ago

📁 agency	Fixing device_id naming for telemetry.	17 days ago
📁 generate_schema	merge	2 months ago
📁 provider	Add GBFS Feed Url to providers.csv (#205)	8 days ago
📄 .gitignore	adding Pipenv	2 months ago
📄 LICENSE	license	9 months ago
📄 README.md	Update - One Year Permit Link (#201)	a month ago
📄 ReleaseGuidelines.md	Adding Release guidelines (#147)	3 months ago
📄 ReleaseNotes.md	updating release notes for 0.2.x	2 months ago
📄 providers.csv	Add GBFS Feed Url to providers.csv (#205)	8 days ago



An official website of the City of Austin

Dockless Mobility Overview

Transportation Data and Performance Hub



Dockless Mobility Overview

Month

January 2019



All Modes

📄 Total Trips

250,389

Updated today at 02:33 PM | [Data](#) 📄

📊 Total Miles

261,090

Updated today at 02:33 PM | [Data](#) 📊

📊 Average Miles

1.04

Updated today at 02:33 PM | [Data](#) 📊

⌚ Average Minutes

12.62

Updated today at 02:33 PM | [Data](#) ⌚

★ Total Devices

8,809

Updated today at 02:33 PM | [Data](#) ★

☎ 311 Service Requests

355

Updated today at 01:52 PM | [Data](#) ☎

Dockless Scooters

⚡ Scooter Trips

249,854

Updated today at 02:33 PM | [Data](#) ⚡

📊 Total Miles

259,826

Updated today at 02:33 PM | [Data](#) 📊

📊 Average Miles

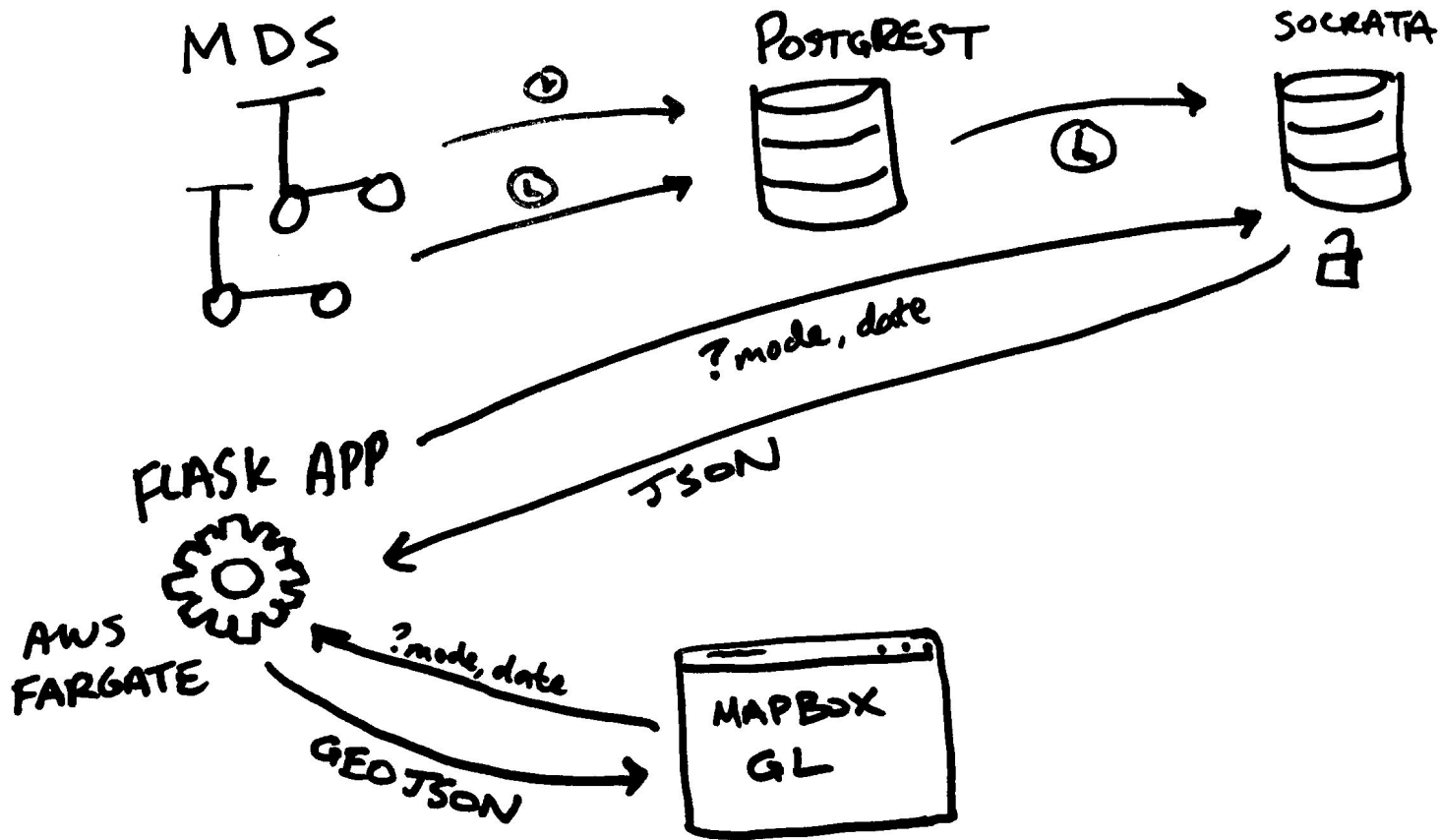
1.04

Updated today at 02:33 PM | [Data](#) 📊

⌚ Average Minutes

12.60

Updated today at 02:33 PM | [Data](#) ⌚



Our Integration Toolkit

[Python Data Publishing](#)

Data publishing scripts

[Docker Data Deploy](#)

Deployment tools

[Knackpy](#)

API wrapper for no-code applications

[Transportation-Data](#)

Open REST interface for data storage/retrieval

[PostgREST Client](#)

Python client to Data publishing scripts

Hands-on

github.com/
cityofaustin/
school-of-data